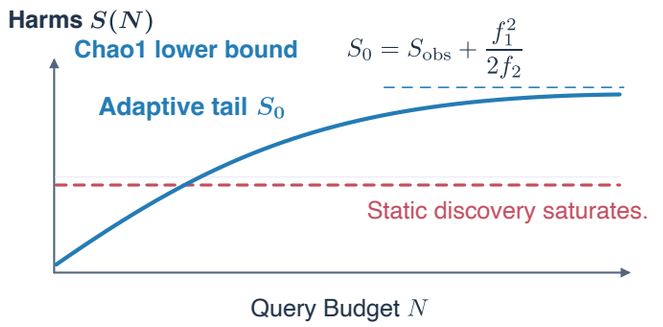
Closed-Loop Research Agenda for Automated Red Teaming

Six open frontiers connect failure discovery to reproducible safety evidence and governed remediation.

Discovery Engine §9.1

Adaptive search estimates mechanism coverage.



Quality-diversity search clusters failure mechanisms.
Singleton and doubleton counts estimate unseen risk.
Coverage anchors include HarmBench and JailbreakBench.

Probes

Decision Oracles §9.2

Search and reporting use separate judgments.

Attack Optimizer
Search uses feedback.

Search Judge J_s
Scores guide search.

Reporting Oracle Panel J_r

- **Classifiers assess safety.** LlamaGuard · WildGuard
- **Assertions check compliance.** Rules and regular expressions
- **Experts resolve disagreements.** Human review

Calibration sets the decision boundary τ.

The search judge guides the optimizer.
Independent oracles arbitrate reported verdicts.
CoinFlip probes evaluator reliability.

Verdicts

Remediation & Retention §9.5

Four factors separate sources of score changes.

- **Target m** Weights and PEFT
- **Attacker a** Mutations
- **Judge j** Rubrics
- **Protocol p** Budget and rules

Regression Replay Suite

- ☑ Versions are recorded.
- ☑ Traces are replayed.
- ☑ Drift is measured.

Utility Guard

Benign MMLU tasks probe over-refusal.

Invariant: $\delta_{\text{drift}} < \epsilon$

Remediation uses DPO, Safe-RLHF and unlearning.
Factor-wise comparisons retain versioned evidence.
BeaverTails and OR-Bench anchor evaluation.

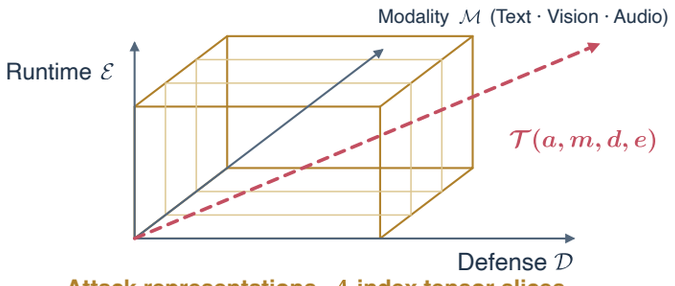
Continuous Closed-Loop Assurance Lifecycle

Evidence connects discovery, repair and disclosure.

Threat Transfer §9.4

Transfer varies across four threat-space axes.

Threat Space: $\mathcal{S} = \mathcal{A} \times \mathcal{M} \times \mathcal{D} \times \mathcal{E}$



Attack representations $\mathcal{A}$ index tensor slices.

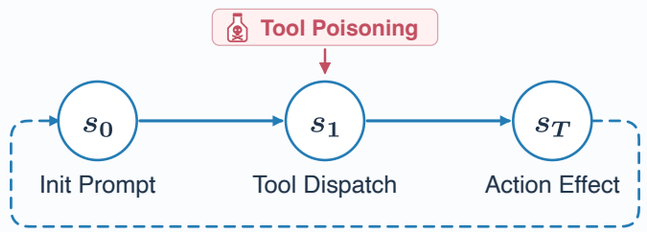
Transfer tests cover low-resource dialects
and culturally specific safety contexts.
Transfer anchors include PoEX, MultiJail and Any-to-Any.

Vectors

Agent Execution §9.3

Sandboxed trajectories support stateful testing.

Execution Sandbox



Snapshot reversion restores recorded state.
Containers and API boundaries isolate execution.
AgentDojo, AgentPoison and OSWorld anchor testing.

Traces

Governance & Disclosure §9.6

Disclosure uses tiered access and coordinated triage.

Evidence Access Tiers

- Aggregate findings are publicly available.
- Vetted users receive gated artifact access.
- Actionable PoC chains remain withheld.

Coordinated Flaw Disclosure (CVD)

Proof is documented. FLARE-AI defines fields.
Vendors triage flaws. Fixes follow quarantine.
Advisories are issued. Release is coordinated.

FLARE-AI and MITRE ATLAS structure reports.
Vendor triage links evidence, fixes and advisories.
Infrastructure includes Inspect AI and CERT/CC.

Governed disclosures and regressions renew coverage seeds.

Schematics are conceptual.